

LAB 1 :

```
package classLab;

public class first {
    Run | Debug
    public static void main(String args[]) {
        System.out.println("Hello world form Eclipse");
        System.out.println("Hi \"Satyam\" \n Roll no  : \t \"2401641530164\"");
        System.out.println("Name : Satyam Chauhan\nBranch : CS-AIML-2A (elite)");
    }
}
```

```
satyam@laptop:~/Desktop/java/src$ java classLab/first
Hello world form Eclipse
Hi "Satyam"
Roll no  :      '2401641530164'
Name : Satyam Chauhan
Branch : CS-AIML-2A (elite)
```

LAB 2 :

```
package classLab;
public class second {
    Run | Debug
    public static void main(String[] args) {
        if(args.length !=2 ){
            System.out.println("Please enter only two numbers");
            return ;
        }
        int a = Integer.parseInt(args[0]);
        int b = Integer.parseInt(args[1]);
        System.out.println("Sum is : "+ (a+b));
        System.out.println("Name : Satyam Chauhan\nBranch : CS-AIML-2A (elite)");
    }
}
```

```
● satyam@laptop:~/Desktop/java/src$ javac classLab/second.java
● satyam@laptop:~/Desktop/java/src$ java classLab/second 18 19
Sum is : 37
Name : Satyam Chauhan
Branch : CS-AIML-2A (elite)
```

LAB 3 :

```
package classLab;

public class third {
    Run | Debug
    public static void main(String[] args) {
        Student s1 = new Student(s: "Satyam",i: 123);
        s1.display();
        Rectangle r1 = new Rectangle(l: 12,b: 12);
        r1.calculateArea();
        System.out.println("Name : Satyam Chauhan");
        System.out.println("Branch : CS-AIML-2A (elite)\nRoll No: 2401641530164");
    }
}
```

```
package classLab;

public class Rectangle {
    int length ;
    int width ;
    Rectangle(int l , int b ){
        this.length = l ;
        this.width = b;
    }
    public void calculateArea() {
        System.out.println("Area of Rectangle : "+this.length*this.width);
    }
}
```

```
package classLab;

public class Student {
    String name ;
    int roll ;
    Student(String s , int i){
        this.name = s;
        this.roll = i;
    }
    public void display(){
        System.out.println("Name : "+this.name +" \nRoll: "+this.roll);
    }
}
```

```
● satyam@laptop:~/Desktop/java/src$ java classLab/third
Name : Satyam
Roll: 123
Area of Rectangle : 144
Name : Satyam Chauhan
Branch : CS-AIML-2A (elite)
Roll No: 2401641530164
```

LAB 4 :

```
package classLab;

public class forth {
    Run | Debug
    public static void main(String args[]) {
        Dog d1 = new Dog(name: "Kalu");
        d1.sound();
        d1.sound(a: "Bhow BHow Bhow");
        System.out.println("Name : Satyam Chauhan");
        System.out.println("Branch : CS-AIML-2A (elite)\nRoll No: 2401641530164");
    }
}
```

```
package classLab;

public class Animal {
    public void sound() {
        System.out.println("Voice of an Animal");
    }
}
```

```
package classLab;

public class Dog extends Animal {
    String name ;
    Dog(String name ){
        this.name = name;
    }
    public void sound(){
        System.out.println("Bark..");
    }
    public void sound(String a){
        System.out.println(a);
    }
}
```

```
● satyam@laptop:~/Desktop/java/src$ java classLab/forth
Bark..
Bhow BHow Bhow
Name : Satyam Chauhan
Branch : CS-AIML-2A (elite)
Roll No: 2401641530164
```

LAB 5 :

```
public class fifth {  
    Run | Debug  
    public static void main(String[] args) {  
        int a = 10 , b = 0;  
        try{  
            System.out.println(a/b);  
        }catch(ArithmeticException e ){  
            System.out.println("Divide by zero error");  
        }finally{  
            System.out.println("Arithmentic program ends here");  
        }  
        MyThread t1 = new MyThread(s: "A");  
        MyThread t2 = new MyThread(s: "B");  
        t1.run();t2.run();  
        System.out.println("Name : Satyam Chauhan");  
        System.out.println("Branch : CS-AIML-2A (elite)\nRoll No: 2401641530164");  
    }  
}
```

```
package classLab;  
  
public class MyThread extends Thread{  
    String name ;  
    MyThread(String s){  
        this.name = s;  
    }  
    public void run(){  
        for(int i = 0 ; i<2 ;i++){  
            System.out.println("I am Thread "+this.name);  
        }  
    }  
}
```

```
● satyam@laptop:~/Desktop/java/src$ java classLab/fifth  
Divide by zero error  
Arithmentic program ends here  
I am Thread A  
I am Thread A  
I am Thread B  
I am Thread B  
Name : Satyam Chauhan  
Branch : CS-AIML-2A (elite)  
Roll No: 2401641530164
```

LAB 6 :

```
package studentInfo;

public class sixth {
    Run | Debug
    public static void main(String[] args) {
        Student s1 = new Student(name: "Satyam Chauhan",
            rollNo: "2401641530164",
            section: "CS_AIML_2C_ELITE",
            phone: "+1+1234567890",
            email: "satyam@gmail.cocm",
            batch: "2024-28");
        s1.display();
    }
}
```

```
package studentInfo;

public class Student {
    String name ;
    String rollNo ;
    String section ;
    String phone ;
    String email ;
    String batch ;
    Student(String name , String rollNo , String section , String phone , String email, String batch ){
        this.name = name ;
        this.rollNo = rollNo ;
        this.section = section;
        this.phone = phone;
        this.email = email ;
        this.batch = batch ;
    }
    void display(){
        System.out.println("Name : "+this.name);
        System.out.println("Roll No : "+this.rollNo);
        System.out.println("Section : "+this.section);
        System.out.println("Phone : "+this.phone);
        System.out.println("Email : "+this.email);
        System.out.println("Batch : "+this.batch);
    }
}
```

```
● satyam@laptop:~/Desktop/java/src$ java studentInfo/sixth
Name : Satyam Chauhan
Roll No : 2401641530164
Section : CS_AIML_2C_ELITE
Phone : +1+1234567890
Email : satyam@gmail.cocm
Batch : 2024-28
```